TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	56.5 ± 3.64
Correlation Stability	87.9 ± 1.22
Distribution Consistency	53.7 ± 6.00
Entropy Preservation	2.3 ± 1.19
Mean Z Score Drift	0.3 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.1 ± 0.30
Data Density	100.0 ± 0.00
Coverage Score	96.6 ± 0.80
Feature Accuracy	70.6 ± 2.37
Novelty Score	0.8 ± 0.02

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	56.707	+2.268	4.2%	OK
sex	0.680	0.876	+0.196	28.9%	Drift
cp	3.158	3.727	+0.569	18.0%	Warn
trestbps	131.690	125.517	-6.173	4.7%	OK
chol	246.693	251.831	+5.138	2.1%	OK
fbs	0.149	0.000	-0.149	100.0%	Drift
restecg	0.990	1.157	+0.167	16.9%	Warn
thalach	149.607	157.157	+7.550	5.0%	OK
exang	0.327	0.169	-0.157	48.1%	Drift
oldpeak	1.040	0.736	-0.304	29.2%	Drift
slope	1.601	1.545	-0.055	3.4%	OK
ca	0.672	0.212	-0.461	68.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	56.5 ± 3.64
Correlation Stability (*2)	87.9 ± 1.22
Distribution Consistency (*3)	53.7 ± 6.00
Entropy Preservation (*4)	2.3 ± 1.19
Mean Z Score Drift (*5)	0.3 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.1 ± 0.30

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	96.6 ± 0.80
Feature Accuracy (*10)	70.6 ± 2.37
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5737704918032788	0.6081967213114754	-1.1102230246251566e-17	
F1	0.6070236204175147	0.5456029086761313	0.5955396078857897	-0.011484012531724852	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5901639344262295	0.5934426229508196	0.5770491803278689	-0.013114754098360637	
F1	0.5480779520963047	0.5238865049532728	0.5237466286219807	-0.024331323474323964	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704919	0.5868852459016394	0.5590163934426229	0.022950819672131133	
F1	0.5231512298313296	0.545160006646472	0.5361787580320171	0.013027528200687543	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.578688524590164	0.5344262295081967	0.6065573770491803	0.027868852459016373	
F1	0.5385134182056105	0.4407972987702033	0.557704246092273	0.01919082788666237	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5901639344262295	0.6032786885245901	0.004918032786885229	
F1	0.5648347697855012	0.5120735284498765	0.5400872239673364	-0.02474754581816473	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.6098360655737705	0.5475409836065575	0.00491803278688524	
F1	0.5198013329305267	0.555505632215588	0.528060431994051	0.00825909906352416	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5868852459016394	0.5327868852459016	-0.02459016393442622	
F1	0.568660979990255	0.5190034317402572	0.54399638392552	-0.02466459606473506	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5262295081967212	0.5442622950819672	0.5278688524590164	0.00163934426229509	
F1	0.5272786716544238	0.5269329341556823	0.529647450947859	0.00236877929343550	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5426229508196722	0.5360655737704917	-0.01803278688524591	
F1	0.44422020714365684	0.404856038293262	0.41459186389765873	-0.02962834324599815	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	61.0	0.0	3.0	118.0	286.0	0.0	0.0	160.0	0.0	0.2754638271089445
2	54.0	1.0	4.0	123.0	201.0	0.0	0.0	167.0	0.0	0.112419484151984
3	53.0	1.0	4.0	133.0	252.0	0.0	2.0	142.0	0.0	0.1384264542002016
4	61.0	1.0	4.0	128.0	314.0	0.0	2.0	162.0	0.0	0.3945372111492038
5	65.0	1.0	4.0	127.0	312.0	0.0	2.0	104.0	1.0	1.75435283893043
6	59.0	1.0	4.0	123.0	253.0	0.0	2.0	158.0	0.0	-0.19353771192939978
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	144.0	283.0	0.0	2.0	167.0	0.0	1.2662053188058144
2	71.0	0.0	4.0	142.0	308.0	0.0	2.0	164.0	0.0	-0.11169312591684293
3	48.0	1.0	3.0	118.0	177.0	0.0	0.0	155.0	0.0	-0.10332612592135192
4	35.0	1.0	3.0	122.0	159.0	0.0	0.0	158.0	0.0	-0.05794569609990689
5	52.0	1.0	4.0	131.0	205.0	0.0	2.0	138.0	0.0	0.8699516099314196
6	59.0	0.0	4.0	140.0	296.0	0.0	2.0	160.0	0.0	0.17662642427222014
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	44.0	0.0	3.0	130.0	271.0	0.0	0.0	144.0	0.0	-0.04803340641746942
2	60.0	1.0	3.0	128.0	229.0	0.0	0.0	161.0	0.0	-0.06597545386527723
3	61.0	1.0	4.0	138.0	285.0	0.0	2.0	144.0	0.0	-0.10828156649907747
4	53.0	1.0	2.0	139.0	222.0	0.0	0.0	162.0	0.0	-0.07304989470819748
5	54.0	0.0	3.0	140.0	227.0	0.0	2.0	168.0	0.0	-0.24380148102421814
6	54.0	1.0	4.0	109.0	248.0	0.0	0.0	88.0	1.0	0.9680754715568822
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	68.0	1.0	3.0	134.0	188.0	0.0	2.0	155.0	0.0	0.2053655151922516
2	68.0	1.0	3.0	136.0	202.0	0.0	2.0	167.0	0.0	-0.031434416703553236
3	50.0	1.0	3.0	118.0	223.0	0.0	0.0	182.0	0.0	-0.09965716514003087
4	55.0	1.0	4.0	130.0	186.0	0.0	0.0	149.0	0.0	0.01532351613903346
5	60.0	1.0	4.0	114.0	284.0	0.0	2.0	122.0	1.0	2.9090441600523627
6	59.0	1.0	3.0	127.0	228.0	0.0	0.0	165.0	0.0	0.24108395433647692
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	52.0	1.0	3.0	131.0	220.0	0.0	0.0	163.0	0.0	-0.012595109899872614
2	63.0	1.0	3.0	122.0	224.0	0.0	0.0	175.0	0.0	0.11658558512836056
3	62.0	1.0	4.0	122.0	208.0	0.0	0.0	171.0	0.0	0.11601014234194316
4	48.0	1.0	3.0	115.0	215.0	0.0	0.0	179.0	0.0	0.23359739699712323
5	58.0	1.0	4.0	131.0	238.0	0.0	2.0	115.0	0.0	2.0356170431081918
6	67.0	1.0	4.0	128.0	232.0	0.0	2.0	154.0	0.0	0.20433043670022732
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	50.0	1.0	3.0	108.0	211.0	0.0	0.0	174.0	0.0	0.010806475724036854
2	56.0	1.0	4.0	119.0	240.0	0.0	0.0	150.0	0.0	0.17240879896048622
3	59.0	1.0	3.0	127.0	278.0	0.0	2.0	161.0	0.0	-0.014012524366691745
4	69.0	1.0	4.0	123.0	252.0	0.0	0.0	103.0	0.0	0.9960624695374967
5	59.0	1.0	4.0	122.0	297.0	0.0	2.0	175.0	0.0	0.056757576693880028
6	62.0	1.0	4.0	106.0	236.0	0.0	0.0	179.0	0.0	0.10305281186981138
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	40.0	1.0	4.0	110.0	261.0	0.0	2.0	181.0	0.0	-0.019997581952286875
2	55.0	1.0	4.0	135.0	241.0	0.0	2.0	158.0	0.0	0.07027855438347476
3	61.0	1.0	4.0	119.0	302.0	0.0	2.0	149.0	0.0	-0.4433504554514075
4	74.0	0.0	4.0	135.0	330.0	0.0	2.0	155.0	0.0	0.016906178102544606
5	50.0	1.0	4.0	119.0	261.0	0.0	0.0	150.0	0.0	0.4797033288114752
6	58.0	1.0	3.0	144.0	190.0	0.0	0.0	157.0	0.0	0.3185597157560021
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	45.0	1.0	2.0	133.0	294.0	0.0	0.0	181.0	0.0	0.09288326725572038
2	66.0	0.0	3.0	149.0	364.0	0.0	2.0	147.0	0.0	1.1465695595709169
3	54.0	0.0	2.0	123.0	262.0	0.0	0.0	174.0	0.0	0.1833224306260182
4	56.0	0.0	4.0	131.0	268.0	0.0	2.0	162.0	0.0	0.9397943387571338
5	66.0	1.0	3.0	131.0	234.0	0.0	0.0	181.0	0.0	0.1381717517443825
6	55.0	1.0	4.0	110.0	208.0	0.0	0.0	125.0	0.0	0.1560875577776407
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	55.0	1.0	4.0	128.0	173.0	0.0	2.0	147.0	1.0	2.855435278301285
2	62.0	1.0	4.0	133.0	220.0	0.0	2.0	149.0	0.0	2.947855241672478
3	63.0	1.0	4.0	148.0	299.0	0.0	2.0	109.0	1.0	2.383369757564906
4	40.0	1.0	4.0	111.0	219.0	0.0	0.0	101.0	0.0	0.042959076172518
5	58.0	1.0	4.0	134.0	299.0	0.0	2.0	163.0	0.0	1.070310417458528
6	54.0	1.0	3.0	111.0	223.0	0.0	2.0	174.0	0.0	0.028120867286011493
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	60.0	1.0	3.0	137.0	312.0	0.0	2.0	159.0	0.0	0.06849332671216773
2	50.0	1.0	4.0	123.0	274.0	0.0	2.0	185.0	0.0	0.061281005130984244
3	62.0	1.0	4.0	138.0	252.0	0.0	0.0	164.0	0.0	-0.38374904608069316
4	57.0	0.0	3.0	107.0	227.0	0.0	2.0	174.0	0.0	0.036426931200548829
5	58.0	0.0	3.0	123.0	216.0	0.0	0.0	155.0	0.0	0.31739872091802657
6	67.0	0.0	3.0	139.0	273.0	0.0	2.0	165.0	0.0	1.4907419768808794
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6